

Machine-Learning-Based Olfactometry: An Auxiliary System for Human Assessors in Olfactory Measurement

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Abstract

Although gas chromatography/olfactometry (GC/O) has been employed as a powerful analytical tool in odor measurement, its application is limited by the variability, subjectivity, and high cost of the trained panelists who are used as detectors in the system. The advancements in data-driven science have made it possible to predict structure-odor-relationship (SOR) and thus to develop machine-learning-based olfactometry (ML-GCO) in which the human panelists may be replaced by machine learning models to obtain the sensory information of GC-separated chemical compounds. However, one challenge remained in ML-GCO is that there are too many odor descriptors (ODs) being used to describe the sensory characteristics of odorants. It is impractical to build a corresponding model for each OD. To solve this issue, we propose a SOR prediction approach based on odor descriptor clustering. 256 representative ODs are firstly classified into 20 categories using a co-occurrence Bayesian embedding model. The categorization effect is explained according to the semantic relationships using a pre-trained Word2Vec model. Various molecular structure features including molecularly parameters, molecular fingerprints, and molecular 2D graphic features extracted by convolutional neural networks, are employed to predict the aforementioned odor categories. High prediction accuracies (Area under ROC curve was 0.800 ± 0.004) demonstrate the rationality of the proposed clustering scenario and molecular feature extraction. This study makes the ML-GCO models much closer to the practical application since they can be expected as either an auxiliary system or complete replacement of human panelists to perform the olfactory evaluation.

44

45 **Keywords:** Gas chromatography/olfactometry, Structure odor relationship, Odor
46 descriptors, Machine learning, Clustering analysis

47 **1. Introduction**

48 Gas chromatography/olfactometry (GC/O) is a key technique that integrates the
49 separation of volatile compounds using a gas chromatograph (GC) with the detection of
50 odor by employing human assessors as olfactometers [1-4]. Contributed by mass
51 spectrometry (MS) analysis and human assessors, GC/O can provide not only the
52 molecular information of complex odor mixtures, but also sensory information of
53 specific odor-active components. Therefore, it has been applied as a critical analytical
54 instrument in various filed, such as food, cosmetics, agriculture, and the environment
55 [5-8].

56 Although human assessors play a very important role, the promotion and
57 application of GC/O measurement in the meantime are also limited by the limitations of
58 human evaluation. The GC/O measurement generally requires multiple people to detect
59 and evaluate volatile compounds by sniffing GC effluent components in real-time [9].
60 Therefore, the subjectivity of panelists at the intra- and inter-individual levels is
61 inevitable. The high cost of training and employing human panelists is also a great
62 problem. In addition, limited by the resolution of the GC instrument as well as the
63 olfactory threshold of human olfaction, the odor-active compounds may be ignored
64 during the GC/O measurement. Recently, many auxiliary software modules, such as
65 AroChemBase (Alpha MOS Co. Ltd.) and GC/MS Off-Flavor Analyzer (Shimadzu Co.
66 Ltd.), have been developed as supplements to GC/O analysis. These auxiliary systems
67 work well in improving the accuracy of the evaluation as well as in reducing the manual
68 labor of human assessors.

69 It is well known that the number of odorants with known descriptors is limited
70 despite having long years of accumulation in the field of flavors and fragrances. The
71 number of odorants with odor descriptors (around 2000) that are registered in those
72 auxiliary systems is far less than that of the total odorant pool (around 44,000) [10].
73 These existing systems may not meet the requirements of GC/O measurement since
74 most of the odor-active compounds still remain unknown to us. This problem may
75 persist for a long time given the difficulty of olfactory measurement. Recently, the use
76 of artificial intelligence (AI) to support analytical purposes has been a potential
77 technology for overcoming the drawbacks of traditional analytical methods [11, 12].

78 The precise relationship between molecular structure and odor perception, termed
79 the structure-odor relationship (SOR), has attracted considerable attention from the
80 aspect of data science [13, 14]. Great efforts have been made in ODs identification
81 based on various molecular features, such as physicochemical parameters, bio-inspired
82 olfactory model, and MS [15]. Moreover, various deep neural network models (DNNs),
83 machine learning (ML) classification frameworks, and topic models have been
84 developed to express the SOR [16-18]. More recently, Debnath and co-workers
85 proposed a model to predict perceptual scent impressions from MS of odorants under an
86 imbalanced dataset [19, 20]. In addition, Mayhew et. al. developed an odorless
87 identification model based on transport features with an extremely high accuracy [21].
88 The abovementioned studies indicated that the SOR would be solved by ML algorithms
89 and chemometrics. Therefore, our research team previously developed a method for
90 conducting ML-based GC/O (ML-GCO), in which olfactometry detection can be done

91 by an ML classifier, thus reducing the need for a human panelist [22].

92 It should be mentioned that, however, there are many problems needed to be
93 overcome before the practical application of ML-GCO. For example, the dimension of
94 odor space remains unknown and it is not yet clear on the primary dimensions of
95 olfactory like vision or gustatory [23]. Odors are highly complex and people are known
96 to disagree regarding their linguistic descriptions of smell sensations. The number of
97 odor compounds is estimated at over 400,000 which are described by several hundred
98 odor descriptors [24]. In most reported work, only a small number of typical, common
99 odor descriptors have been included, leaving the majority of them undefined [22, 25,
100 26].

101 Many researchers have demonstrated that odor descriptors are inherently connected
102 in the odor structure space. For example, carbon-chain length and functional groups
103 have been confirmed as critical factors in odor perception [27-29]. This means the odor
104 descriptors can be clustered according to their similarity evaluations and thus cover the
105 odor descriptor space as many as possible. One challenge in the prediction of ODs is
106 defining the primary dimensions of the olfactory perceptual space [30]. In addition,
107 odors are highly complex, and people are known to disagree regarding their linguistic
108 descriptions of smell sensations [31-33]. Although most ODs can be predicted with high
109 precision, infrequent smells, such as apricot and chocolate, were shown to difficult to be
110 predicted [25]. Yet, previous studies have demonstrated that odor descriptors are
111 inherently connected in perceptual space [26]. It indicated that co-occurrence of ODs
112 from odor database would be a reasonable direction to find odor categories. Recently,

113 Pandey and co-workers proposed a vibration-based biomimetic odor classification
114 method based on the vibrational spectrum of an odorant molecule [34]. However, only
115 20 odorant molecules were considered for analyzing odor perceptual classes, which
116 would not enough to offer insight into the olfaction space. Therefore, this issue could be
117 addressed via a rational relationship for OD similarity evaluations and clustering using a
118 large odor database.

119 In response to the above problems, ODs embedding and clustering approaches are
120 proposed. A schematic of the data-processing method is illustrated in Fig. 1. ODs from
121 different data sources are collected and merged, and Bayesian co-occurrence embedding
122 (BCE) was employed for odorants and ODs embedding. Based on the distance between
123 odorant vectors and OD vectors, ODs can be calibrated and relabeled. And then
124 hierarchical clustering analysis (HCA) was performed on the embedded OD vectors to
125 investigate the internal relationship between ODs and their clustering result is discussed.
126 We show that a total of 256 ODs were clustered in 20 categories, and most of the
127 categories can be supported by previous research. In addition, the structure parameters
128 of odorant molecules are employed for training ML models to verify the rationality of
129 the proposed clustering scenario. Results show that the smell categories can be
130 predicted by the models established by molecularly structure features of odorants
131 successfully (Area under ROC curve: 0.800 ± 0.004 , precision: 0.595 ± 0.004 , recall:
132 0.721 ± 0.003 and F-score: 0.570 ± 0.007 , $p < 0.0001$ Wilcoxon test). It indicated that the
133 smell clusters proposed in this study are supported by the SOR. The proposed odor
134 category is not only expected as a novel research direction for developing ML-GCO,

135 but also applies a reference for understanding the biology of olfaction. This study makes
136 the ML-GCO models much closer to the practical application since they can be
137 expected as either an auxiliary system or complete replacement of human panelists to
138 perform the olfactory evaluation.

139 **2. Material and methods**

140 **2.1 Data preparation**

141 To understand the internal relationships between smell percepts, we collected the
142 chemical abstracts service (CAS) data of odorants and their odor perceptions via both
143 web scraping and manual methods. We used three publically available odor databases,
144 including the Flavors and Fragrances database (Sigma-Aldrich) [35], the Good Scents
145 database [36], and the Odor Map database [37]. Detailed information on odor databases
146 used in the present study is summarized in Table 1. Based on CAS number, the
147 simplified molecular-input line-entry system (SMILES) data were collected from
148 PubChem (<https://pubchem.ncbi.nlm.nih.gov>) [38]. Using RDKit software
149 (<http://www.rdkit.org>) [39], molecular structure images, molecular parameters, and
150 molecular fingerprints were obtained for further analysis. In the present study,
151 chemicals without odor descriptors or those considered "odorless" were not considered.
152 Finally, 2849 molecules with 510 ODs were collected and analyzed. All of the ODs in
153 the database are listed in Table S1.

154 **2.2 Bayesian co-occurrence embedding**

155 To normalize the labels for odorants from all databases, BCE was introduced for
156 OD vector embedding [40]. A brief description of BCE is illustrated in Fig. 2. Based on

BCE, a preference matrix for each odorant can be generated. A plus sign (+) indicates comparisons with OD_d , OD_j is the label for the odorant_i, a minus sign (-) indicates the opposite, and a question mark (?) indicates the unknown value that is estimated by the BCE algorithm. Therefore, the individual probability that an odorant prefers OD_d to OD_j was defined as:

$$P(d >_c j | \Theta) := \sigma(\hat{x}_{cdj}(\Theta)) \quad (1)$$

where σ is the logistic sigmoid function, Θ is the parameters of the BCE model, and $\hat{x}_{cdj}(\Theta)$ is the score indicating the degree to which chemical C prefers OD_d rather than OD_j . According to maximum posterior estimation, the generic optimization criterion for each odorant could be estimated as follows:

$$\begin{aligned} \text{BCE Loss} &:= \ln p(\Theta | >_c) \\ &= \ln p(>_c | \Theta) P(\Theta) \\ &= \ln \prod_{(c,d,j) \in D_s} \sigma(\hat{x}_{cdj}) P(\Theta) \\ &= \sum_{(c,d,j) \in D_s} \ln \sigma(\hat{x}_{cdj}) + \ln P(\Theta) \\ &= \sum_{(c,d,j) \in D_s} \ln \sigma(\hat{x}_{cdj}) - \lambda_{\Theta} \|\Theta\|^2 \end{aligned} \quad (2)$$

where λ_{Θ} are model-specific regularization parameters. Therefore, the gradient of the loss function with respect to the model parameters is:

$$\begin{aligned} \frac{\partial \text{BCE Loss}}{\partial \Theta} &= \sum_{(c,d,j) \in D_s} \frac{\partial}{\partial \Theta} \ln \sigma(\hat{x}_{cdj}) - \lambda_{\Theta} \frac{\partial}{\partial \Theta} \|\Theta\|^2 \\ &\propto \sum_{(c,d,j) \in D_s} \frac{-\exp(-\hat{x}_{cdj})}{1 + \exp(-\hat{x}_{cdj})} \cdot \frac{\partial}{\partial \Theta} \hat{x}_{cdj} - \lambda_{\Theta} \Theta \end{aligned} \quad (3)$$

Finally, the model parameters (Θ) could be updated using the assigned learning rate η and stochastic gradient descent as follows.

$$\Theta \leftarrow \Theta - \eta \left(\frac{-\exp(\hat{x}_{cdj})}{1 + \exp(-\hat{x}_{cdj})} \cdot \frac{\partial}{\partial \Theta} \hat{x}_{cdj} - \lambda_{\Theta} \Theta \right) \quad (4)$$

174 Unlike k-nearest neighbor (kNN) collaborative filtering or matrix factorization
175 (MF), BCE applies a Bayesian optimization criterion to generate odorant similarity
176 rankings based on pairs of ODs (i.e. the odorant-specific order of two ODs). As an
177 offline embedding method, Bayesian optimization has advantages over the standard
178 learning techniques for MF and kNN [41, 42]. In the present study, the relabeling results
179 produced by BCE were evaluated via the normalized discounted cumulative gain
180 (NDCG).

$$181 \quad NDCG_p = \frac{DCG_p}{IDCG_p} \quad (5)$$

182 where the discounted cumulative gain (DCG) and ideal discounted cumulative gain
183 (IDCG) were calculated as:

$$184 \quad \begin{aligned} DCG_p &= \sum_{i=1}^p \frac{2^{rel_i} - 1}{\log_2(i+1)} \\ IDCG_p &= \sum_{i=1}^{REL_p} \frac{2^{rel_i} - 1}{\log_2(i+1)} \end{aligned} \quad (6)$$

185 2.3 Molecular graphic feature extraction

186 To extract the necessary numerical features from the molecular structure images,
187 we considered four types of pretrained convolutional neural network (CNN)
188 frameworks, including VGG-16, Resnet, Densnet, and Alexnet. Detailed information
189 for these CNNs is given in these papers [43-46]. In the present study, the CNNs were
190 only applied as molecular-graphic feature extractors for the SOR model calibration.

191 2.4 SOR model calibration

192 The dataset for the odor category prediction was a typical imbalanced data set
193 because the class distribution of the positive samples (minor samples with specified

194 odor categories) and negative samples (major samples with non-specified odor
195 categories) was not uniform. Inspired by Mordelet's research, we considered bagging
196 classifiers to be a feasible method for learning with an imbalanced data set [47]. Detail
197 description of transductive bagging learning is presented in the support information. In
198 the present study, the sample pool was divided into training and test sets with a 3:1 ratio
199 via random stratified sampling. In addition, the number of bootstraps was set as 100 and
200 the subsample number was the same as that for the positive numbers. Considering the
201 sample size, we employed GBDT and GLVQ to predict the odor clusters in the present
202 study [48, 49]. More introduction to these models is described in the support
203 information. Finally, the optimal feature extractor and model combination was
204 determined by considering the area under the ROC curve (AUC-ROC), precision, recall,
205 and F1-score of the test set, respectively. Detailed information for these metrics is
206 presented in the support information materials. In the present study, data and models
207 were processed and analyzed using Python (ver. 3.9.0) and R (ver. 4.1.1).

208 **2.5 Word2Vec embedding model**

209 To investigate the semantic internal relationships between ODs for each odor
210 category, we used Google's pre-trained Word2Vec model to create semantic
211 presentations for ODs. The model, which contains 300-dimensional vectors for 3
212 million words and phrases, was trained using the Google News dataset. Two types of
213 model architectures, including the continuous bag-of-words (CBOW) model and the
214 continuous skip-gram model, were developed for learning latent presentations for words.
215 More detailed information can be found here [50]. The proposed model has been

confirmed to perform better than previous techniques based on different types of neural networks. In addition, the proposed vectors provide the state-of-the-art performance for measuring semantic word similarities.

3. Results and Discussion

3.1 Odor perception embedding and calibration

In the present study, the idea of co-occurrence in BCE was introduced for odor perception calibration. Using the BCE method, ODs can be embedded as numerical vectors. According to the cosine similarity between the odorant and OD vectors, the top 20 nearest ODs were considered as candidates for each odorant. As a critical factor, optimization of the embedded dimension is an important consideration. The normalized discounted cumulative gain for the top 20 ODs (NDCG@20) under different embedded dimensions is shown in Fig. S1. The NDCG@20 increased with the number of embedded dimensions. Embedded vectors with 64 dimensions performed significantly better than embedded vectors with 32, 16, and 8 dimensions ($p < 0.001$), and were not significantly different from embedded vectors with 128, 256, 512 dimensions. Considering accuracy and computational efficiency, we selected 64 as the optimal number of embedded dimensions for the BCE in the present study. The distribution of ODs before and after calibration is shown in Fig. S2, and detailed information is given in Table S1. In summary, the sample size of the ODs increased after calibration in 70.58 % of cases. In this type of statistics, the sample size should always be more than 20. Using the BCE algorithm, the sample size increased to more than 20 in 61 ODs (11.96 %), which were then considered for further analysis. Finally, 265 ODs (51.96 %)

238 were selected for afterward clustering analysis in the present study.

239 **3.2 Clustering characterization for odor descriptors**

240 To quantify the inner relationships between ODs, we performed hierarchical
 241 clustering based on the Euclidean distances between the embedded vectors of smell
 242 descriptors. Because the basis of olfaction has not yet been established, we turned to
 243 previous research to explain our clustering results. The well-known Dravnieks [51] and
 244 DREAM [52] datasets include 19 types of descriptors: the scent of a bakery, sweet, fruit,
 245 fish, garlic, spices, cold, sour, burnt, acid, warm, musky, sweaty, ammonia/urinous,
 246 decayed, wood, grass, floral, and chemical [53]. Based on an analysis of previous
 247 research [54, 55], we considered 20 to be a reasonable number of clusters. The results
 248 were organized and depicted using dendrograms, as shown in Fig. 3 and Fig. S3, S4. In
 249 the clustering results, the descriptors with semantic similarity were almost always
 250 grouped in the same class. Specifically, cluster-1 was composed of sweaty-like and
 251 fish-like descriptors, which were considered unpleasant odors. ODs related to the scent
 252 of a bakery were clustered in cluster-2, and burnt-like descriptors were present in
 253 cluster-4 and cluster-12. Most groups, including cluster-5 (milky-like), cluster-6
 254 (spicy-like), cluster-7 (musky or green-like), cluster-10 (wine or ester-like), cluster-11
 255 (fruit acid-like), cluster-13 (cold or fresh-like), cluster-14 (garlic-like), cluster-15
 256 (fruit-like), cluster-16 (floral), cluster-17 (cold-like), cluster-18 (musky and wood-like),
 257 and cluster-20 (wood-like), were supported by the core smell descriptors proposed in
 258 previous research [56].

259 Although most of the descriptors with similar linguistic meanings were clustered in

the same group, special cases were also observed in some clusters. For cluster-3, the ODs included fruity, sweet, floral, etc., and so we regarded this as the sweet or fruit-like group. However, some descriptors related to plants, such as spicy, woody, herbal, and green, were also present in cluster-3. In addition, we found tomato, chrysanthemum, rotten cabbage, and radish in cluster-14, and labeled this as the garlic-like group. To investigate the reasons for this "mis-clustering" of smell descriptors, the odorants contained in these ODs were extracted and analyzed. For example, the odor of methyl mercaptan (CAS: 74-93-1) is reminiscent of garlic or rotten cabbage, while Erucin (CAS: 4430-36-8) and Berteroin (CAS: 4430-42-6) are labeled as the odors of cabbage and radish. The link between "garlic" and "rotten cabbage", and between "rotten cabbage" and "radish" indicated that these three descriptors could be clustered together, which can explain the presence of cabbage and radish in garlic-like clusters. In addition, Methyl propyl disulfide (CAS: 2179-60-4) is labeled as "radish", "mustard", "tomato", "garlic", etc., and Ethyl methyl sulfide (CAS: 2179-60-4) is labeled as "garlic", "tomato", "rotten cabbage", etc. Thus, the odor semantic database showed an internal connection between the smell of tomato and that of garlic, which could be explained by the latent relationships between the ODs. The above-mentioned analyses demonstrated that semantic descriptor clustering based on embedded vectors using the BCE method is a reasonable approach for understanding internal affiliations. Even though most of the clusters could be explained by common sense or previous research, some clusters, such as cluster-8, were not found to belong to any 'well-known' smell categories proposed by previous studies. In cluster-8, descriptors such as tea, root, basil, thyme, and buchu

could be regarded as woody or herbal. Furthermore, fruit-like (mango, blueberry, passion fruit) and mild descriptors (faint, bland, slightly) were also clustered here. The plants associated with these ODs, such as tea or basil, are mild and herbaceous, which could explain this clustering result. In cluster-10, most of the descriptors were related to alcohol, such as wine-like, powerful, rum-like, alcoholic, etc. Wine flavor descriptors, such as berry, juicy, jam, candy, ether, and ester, were also identified. Consequently, cluster-10 was considered to have the aroma of wine. Chemical-like descriptors, including oil, chemical, and fusel, were also clustered here, which indicated that cluster-10 comprised multiple types of smell perceptions. Interestingly, unpleasant smell descriptors, such as sweaty/fish-like (cluster-1) and garlic-like (cluster-14), could be easily clustered, which confirmed that flavor-like impressions were regarded as more complex than unpleasant perceptions.

3.3 Mapping odor descriptors in t-SNE space

To investigate the internal relationships between odor categories, we visualized the data using Barnes-Hut t-distributed stochastic neighbor embedding (t-SNE) as an unsupervised low dimension presentation method. Detail explanation of the t-SNE method is described in support information. As illustrated in Fig. 4, odor descriptors from 20 categories were mapped in t-SNE space via manifold embedding (an interactive version is at <https://shangliang0225.shinyapps.io/molodorperception>). The details for the embedded data calculated using t-SNE are presented in Table S2. The analysis illustrated that ODs from cluster-1 (sweaty or fish-like), cluster-2 (bakery-like), cluster-3 (woody or herb-like), cluster-7 (musky or green-like), cluster-11 (fruit or

acid-like), cluster-12 (burnt-like), and cluster-14 (garlic-like) were clustered together. Additionally, descriptors from cluster-4 (burnt-like), cluster-13 (cold or fresh-like), cluster-16 (floral), and cluster-20 (wood-like) were clustered in multiple groups. For cluster-4, we found that part-1 (including chicken, lamb, and savory) was on the left of the t-SNE map, and part-2 (including nutty, meaty, sulfurous, coffee, burnt, roasted, cooked, and meat) was on the right of part-1. In addition, we found that beef was located near part-1 of cluster-4, which could be explained by the meat-like smell of beef. Smoky was located near part-2 of cluster-4, which was close to perceptions of burnt or roasted. In summary, plants or herb-related smell perception categories, such as cluster-6, cluster-7, cluster-8, cluster-13, and cluster-19, were located at the top area of the t-SNE space. In addition, fruity or alcohol-like categories (including cluster-10, cluster-11, and cluster-15) were neighbors, and were below the plant or herb-related categories. This demonstrates that the relationships between categories can be illustrated in t-SNE space, and that their locations can also be explained by their semantic meanings.

3.4 Comparison with Word2Vec semantic embeddings

Odor descriptors are not only used to express odor feelings, but are also applied in daily written communication. To explore the pure semantic relationships for each category, we conducted a linguistic analysis. We employed a pre-trained Word2Vec model provided by Google as a feature extractor to generate word vectors containing 3 million words based on roughly 100 billion words from a Google News dataset. Correlation box plots for odor categories are given in Fig. 5 and Table S3. Correlation

heat maps and their distributions are presented in Fig. S5 and Fig. S6. According to the analysis, cluster-14 (garlic-like, 0.479 ± 0.128 , $p < 0.0001$) and cluster-6 (spicy-like, 0.443 ± 0.0674 , $p < 0.001$) had higher internal correlations than the other odor categories, which indicated that these unpleasant perceptions had closer internal relationships in Word2Vec space. In contrast, some odor categories, such as cluster-10 (0.187 ± 0.136 , $p < 0.0001$), cluster-8 (0.213 ± 0.165 , $p < 0.01$), cluster-9 (0.217 ± 0.139 , $p < 0.05$), and cluster-19 (0.227 ± 0.174 , $p < 0.0001$), had lower correlations than the others. As mentioned above, cluster-10, cluster-8, and cluster-19 could not be defined as any 'well-known' smell categories. Furthermore, cluster-9 was composed of multi-odor perception categories, such as herb-like and chemical-like. However, the correlations between most odor categories were lower than 0.6, which demonstrates that the co-occurrence of terms in the text was not exactly the same as that of odor semantic descriptors.

3.5 Distribution of odor perception category labels

Before calibrating the odor category identification model, we first investigated the distribution of the samples. The sample distribution for each odor category is shown in Fig. S7a, which indicates that the sample sizes for the odor descriptors were clearly distinct and imbalanced. Fig. S7b illustrates the statistical distribution for the number of clusters of odorants. It demonstrates that most of the odorants belonged to more than two clusters, which can be explained by the complexity and ambiguity of odor perception.

3.6 Odor perception category identification models

348 To verify the rationality of the proposed clustering scenario, we used the molecular
349 structure features to predict the aforementioned odor categories. To assess the potential
350 of molecular feature extraction via four types of pre-trained CNN models, we used
351 molecular parameters (MPs) and molecular fingerprints (FPs) to identify the clusters for
352 odorants, applied GLVQ, and GBDT classification methods, and then compared and
353 evaluated the results.

354 To calibrate the GLVQ models, the maximum number of training iterations, and
355 the number of prototypes per class were set to 5000 and 10, respectively. The overall
356 AUC, precision, recall, and F-score of the GLVQ models under the features extracted by
357 CNNs, molecular parameters, and molecular fingerprint data sets are shown in Fig. 6,
358 and the detailed predictions of the accuracies for each odor cluster are presented in Fig.
359 S8 and Table 2. For cluster-2, cluster-13, and cluster-15, the VGG produced better
360 results than the other feature extraction methods. However, the features extracted by
361 Restnet did a better job in identifying most clusters. In summary, the Restnet model
362 produced a significantly better average AUC (0.742 ± 0.006), precision (0.580 ± 0.003),
363 recall (0.691 ± 0.005), and F-score (0.548 ± 0.009) than the other models ($p < 0.001$).

364 For the GBDT models, the parameters including the learning rate, max depth of
365 trees, the fraction of features for each tree, gamma, min child weight, and subsample
366 values were set to 0.2, 4, 0.5, 2, 0.5, and 0.5, respectively. As illustrated in Fig. S9, the
367 features extracted using GBDT with VGG showed a higher prediction accuracy for
368 cluster-1, cluster-5, and cluster-17 than for the other clusters. In addition, Restnet did a
369 better job of identifying cluster-7, cluster-8, cluster-10, and cluster-18. However,

Densenet showed better prediction performance for most clusters. The average odor cluster identification results calibrated using the GBDT models are shown in Fig. 6 and Table 2, which shows that the identification accuracy of Densenet (AUC 0.800 ± 0.004 , precision 0.595 ± 0.004 , recall 0.721 ± 0.003 , and F-score 0.570 ± 0.007) was significantly higher than that of the other molecular feature extraction datasets ($p < 0.001$).

3.7 SOR model comparison

When we compared the two modeling methods, we found that the GBDT had better identification accuracy than the GLVQ. In general, the GBDT run with features extracted via Densenet had the best identification performance (AUC 0.800 ± 0.004 , precision 0.595 ± 0.004 , recall 0.721 ± 0.003 , and F-score 0.570 ± 0.007 , $p < 0.001$), followed by the Resnet-GBDT (AUC 0.790 ± 0.004 , precision 0.591 ± 0.004 , recall 0.719 ± 0.004 , and F-score 0.563 ± 0.007 , $p < 0.001$) and Alexnet-GBDT (AUC 0.788 ± 0.005 , precision 0.589 ± 0.004 , recall 0.706 ± 0.004 , and F-score 0.562 ± 0.007 , $p < 0.001$). Models trained using features extracted from molecular structure images showed higher identification accuracy (AUC 0.756 ± 0.005 , precision 0.580 ± 0.003 , recall 0.688 ± 0.004 , and F-score 0.544 ± 0.008 , $p < 0.001$) than that calibrated using molecular parameter features (AUC 0.522 ± 0.003 , precision 0.506 ± 0.002 , recall 0.512 ± 0.002 , and F-score 0.407 ± 0.004 , $p < 0.001$). This indicates that molecular spatial structure is more highly correlated with odor perception than pure molecular parameters, which has been previously confirmed by biology experiments [57]. A similar conclusion has been noted in related work [25]. In summary, we suggest that the GBDT run with features extracted by Densenet from molecular structure images is the optimal model

392 for identifying perception clusters of odorants.

393

394 **3.8 Discussion**

395 This paper reported an odor descriptors clustering approach aimed at testing the
396 feasibility of defining odor categories based on the co-occurrence between odor
397 perceptions. We employed the BCE for describing co-occurrence relations between ODs
398 from three odor databases. Results indicated that 256 ODs were clustered into 20
399 categories by HCA. Furthermore, proposed odor categories were supported by not only
400 semantic analysis, but also the SOR. Marcelo et. al. reviewed the extant biological
401 knowledge on olfaction to clarify the dimensionality of smell [58]. They suggested that
402 although the human nose has over 400 olfactory receptors, the dimensionality of odor
403 perception would be around 20 or less. Therefore, 20 odor categories proposed in the
404 present study would be reasonable to understand the biology olfaction perception space.

405 For developing ML-GCO, we can train a model for odor category identification
406 instead of OD identification, which would be easier to train. In the future, a reliable
407 enough model would be developed as an odor perception recommendation system for
408 the assessors of GC/O to reduce their burdens.

409 While our approach yielded OD clustering results based on the co-occurrence
410 relationships, several limitations of the proposed approach should be discussed. First,
411 abundant ODs, such as sweet and fruity, were not considered because of their
412 ambiguities. Therefore, we need to define a metric to remove these ambiguous ODs and
413 to identify characteristic ODs, such as coffee and vanilla. Second, the SOR model

414 should be extended to include mixtures of odorants. Molecular structures cannot be
415 arbitrarily blended by naive linear superposition. Thus, novel approaches and
416 algorithms, such as molecular 3-dimensional interaction embedding, topology graph
417 representations, and mixed MS analysis should be considered for use in extracting
418 critical features for odor mixture presentation. We hope that, with these possible
419 directions, our work will provide a foundation for understanding human olfaction space
420 and finding the basis of odor perception.

421 **4. Conclusion**

422 In this paper, we describe a method for extracting generalized smell perceptions,
423 termed odor categories. For clustering analysis, a metric should be defined to represent
424 the relationship between these perceptions. In this study, we introduced BCE as a vector
425 embedder for describing the co-occurrence relationships between ODs. The ODs of
426 odorants were also calibrated based on the above-mentioned embedded vectors to
427 reduce the diversity evaluation criterion in the databases. After removing infrequent
428 ODs, cluster analyses were performed on the embedded vectors of the ODs. The results
429 indicated that the ODs were clustered in 20 groups, and most of the ODs with similar
430 semantic meanings were clustered in the same class.

431 To verify the rationality of the proposed clustering scenario, we used molecular
432 structure features to predict the aforementioned odor categories. The high prediction
433 accuracy (AUC is 0.8) of the SOR models demonstrated that co-occurrence-based
434 embedded vectors may be feasible descriptors for expressing the similarity between
435 ODs. In addition, our data demonstrate that molecular structures combined with ML

436 methods can be adopted for odor perception cluster identification, which would be an
437 intelligent auxiliary system for existing GC/O.

438 **CRedit authorship contribution statement**

439 **Liang Shang:** Experiments, Data curation, Writing - original draft. **Chunjun Liu:**
440 Supervision, Project integration, Writing - original draft. **Fengzhen Tang:** Supervision,
441 Project integration. **Bin Chen:** Experiments, Data curation. **Lianqing Liu:** Supervision.
442 **Kenshi Hayashi:** Supervision.

443 **Declaration of competing interest**

444 The authors declare that they have no known competing financial interests or
445 personal relationships that could have appeared to influence the work reported in this
446 paper.

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452 **Appendix A. Supplementary data**

453 Supplementary data to this article can be found online at [http:// doi.org/xxx](http://doi.org/xxx). Detail
454 description of algorithms and models, includes t-SNE, transductive bagging learning,
455 model evaluation metrics, GLVQ, GDBT, supplementary figures, and tables for detailed
456 experiment and analysis results of this work (PDF).

457

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590 **Figure captions**

591 Fig. 1. Data processing diagram of odor descriptor clustering analysis.

592 Fig. 2. Odor descriptor embedding and calibration using the Bayesian co-occurrence embedding
593 (BCE) method.

594 Fig. 3. A summary of odor descriptor clustering based on the similarities between BCE
595 embedded vectors.

596 Fig. 4. Odor descriptor clustering generated in the SOR space using the t-SNE method. An
597 interactive version is available at <https://shangliang0225.shinyapps.io/molodorperception>.

598 Fig. 5. Correlation box plots of odor descriptor Word2Vec embeddings from odor perception
599 categories. Results were evaluated using the nonparametric Wilcoxon signed-rank test.

600 Fig. 6. Comparison of average identification area under the ROC (a), precision (b), recall (c),
601 and F-score (d) for the GLVQ and GBDT models under molecular graphic feature
602 extractions, molecular parameters, and molecular fingerprints. Results were evaluated
603 using the nonparametric Wilcoxon signed-rank test.

604

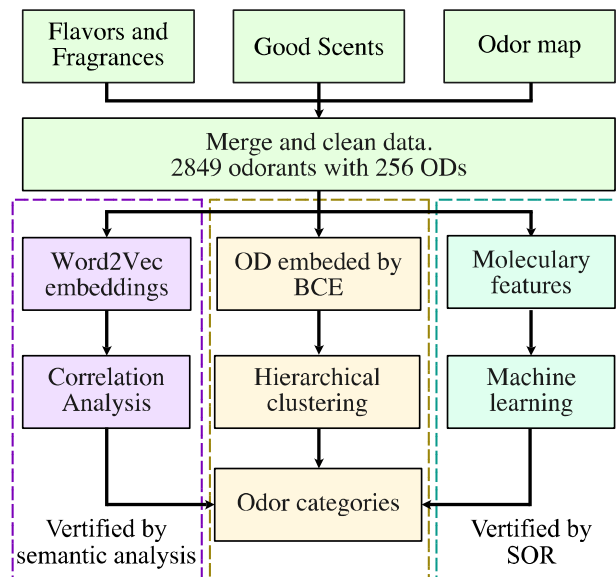


Fig. 1.

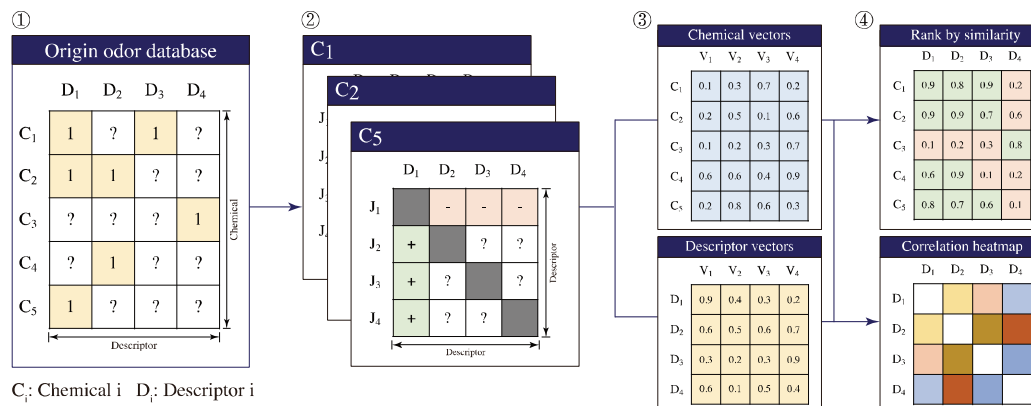


Fig. 2.

Cluster-1 butter; fishy; ammoniacal; cheese; sour, rancid; sweaty; cheesy	Cluster-2 caramel; maple; sugar; fenugreek; bready	Cluster-3 fruity; sweet; fatty; floral; citrus; rose; spicy; woody; green; clean; waxy; herbal; honey; fresh; aldehydic; soapy	Cluster-4 nutty; meaty; sulfurous; coffee; burnt; roasted; cooked; meat; chicken; savory; lamb	Cluster-5 vanilla; powdery; balsam; cherry; almond; bitter; buttery; mild; creamy; dairy; milky; caramellie	Cluster-6 hawthorn; anise; anise; filae; licorice; sassafras; fennel; wintergreen; mimosa	Cluster-7 balsamic; mint; aromatic; cinamyl; pine; camphor; incense; resinous; medicinal; terpenic; fir; noddic; labdanum; thujone
Cluster-8 faint; bland; tea; black; parsley; chamomile; rooty; slightly; mango; basil; bois; passion; buchu; catty; blueberry; thyme	Cluster-9 plastic; strong; sharp; grassy; narcissus; cortex; leafy; musty; earthy; hyacinth; spice; foliage; mushroom; geranium; vegetable; red; metallic; orchid; weedy; galbanum; bean; cumini; mossy; pepper	Cluster-10 berry; juicy; winey; tropical; powerful; raspberry; strawberry; cognac; rum; brown; ether; estery; oil; chemical; alcohol; fusch; ripe; alcoholic; jam; candy	Cluster-11 grapefruit; orange; petitgrain; lemon; lavender; peel; bergamot; sage; lime; mandarin; clary; blossom; neroli; cologne; tangerine	Cluster-12 walnut; fermented; skin; chocolate; cocoa; peanut; potato; nut; popcorn; beef; corn; chip; baked; hazelnut; toasted; grain	Cluster-13 phenolic; solvent; herbaceous; minty; camphoreous; rosemary; medicinal; camphoraceous; smoky; cool; celery; caraway; mentholic; spearmint; peppery; terpene; eucalyptus; peppermint; cooing	Cluster-14 alliacious; onion; garlic; horseradish; sulfury; tomato; chrysanthemum; pungent; cabbage; mustard; radish
Cluster-15 ethereal; wine; apricot; plum; banana; peach; pineapple; apple; pear; wine-like; grape; brandy			Cluster-18 dry; musk; civet; animal; cedar; sandalwood; amber; greasy; vetiver; dusty; ambergris; leather; wood; patchouli; acetate; old	Cluster-19 warm; cinnamon; cassia; deep; clove; camation; root; bay; ginger		Cluster-20 tobacco; coumarin; coconut; tonka; lactonic; huy
Cluster-16 oily; rhubarb; natural; jasmine; lily; jasmine; valley; ylang; petal; ambrette; flower; gardenia; magnolia; muguet; tuberosc; orangeflower; seed		Cluster-17 violet; cucumber; orris; leaf; melon; watery; rind; ozone; marine				

Fig. 3.

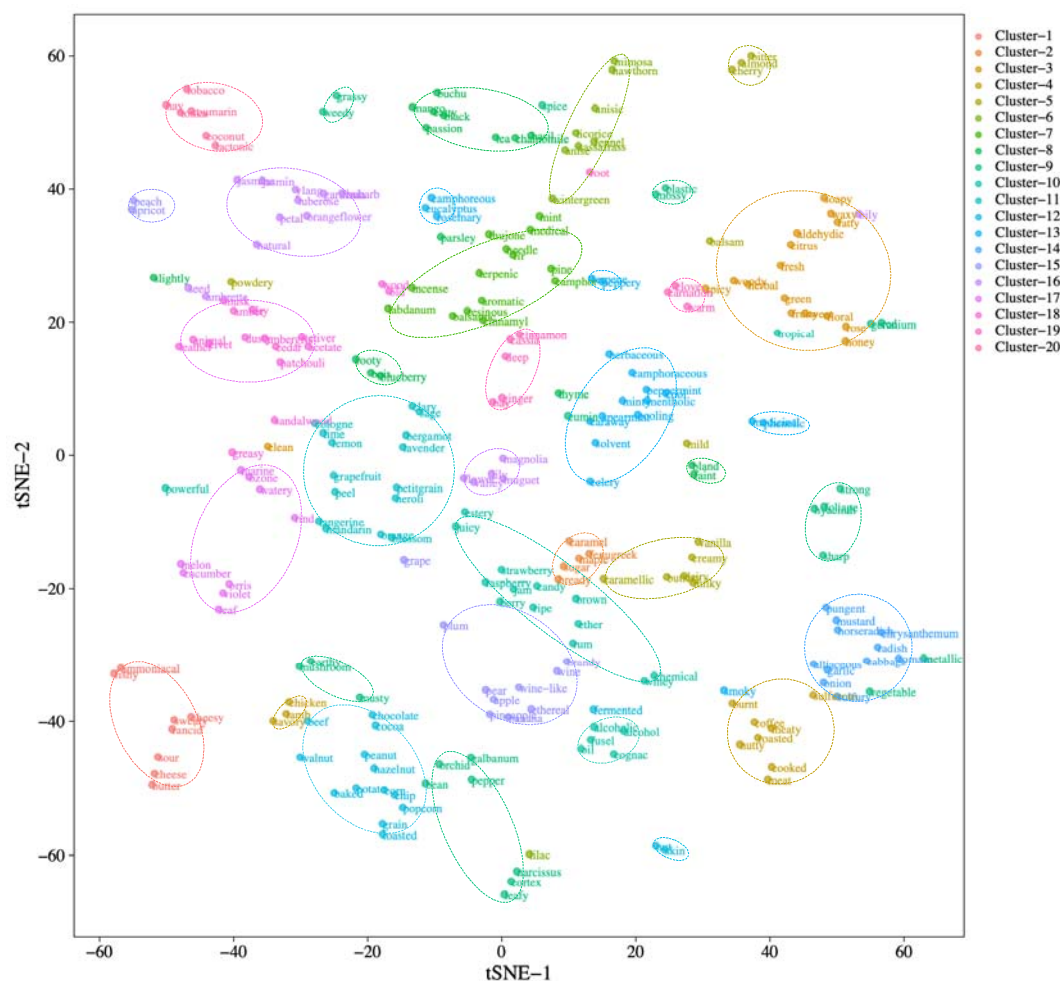


Fig. 4.

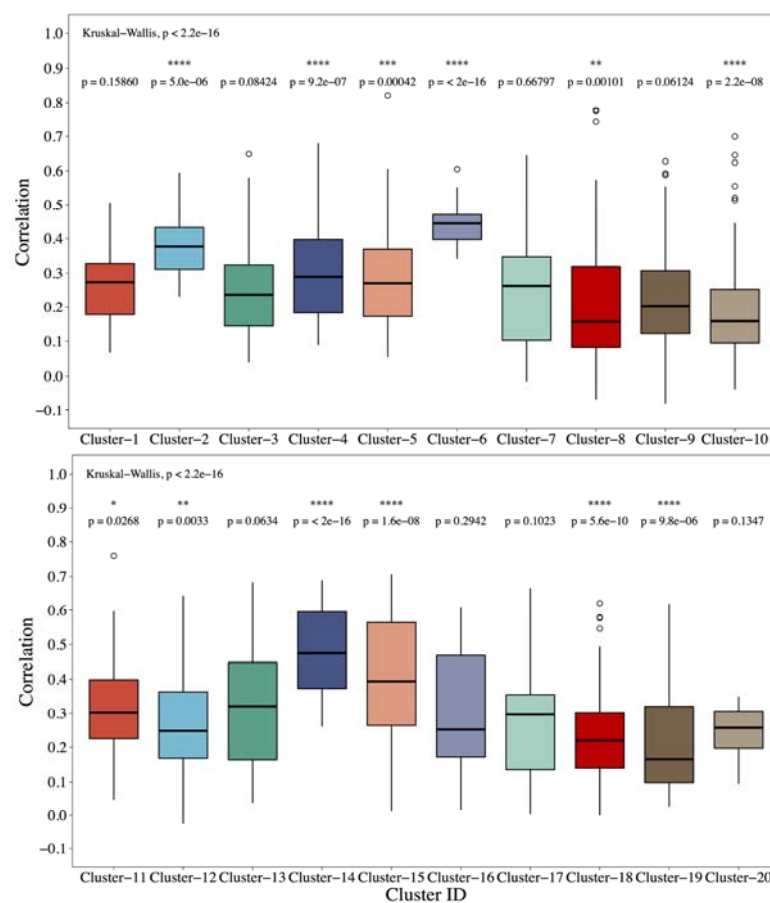


Fig. 5.

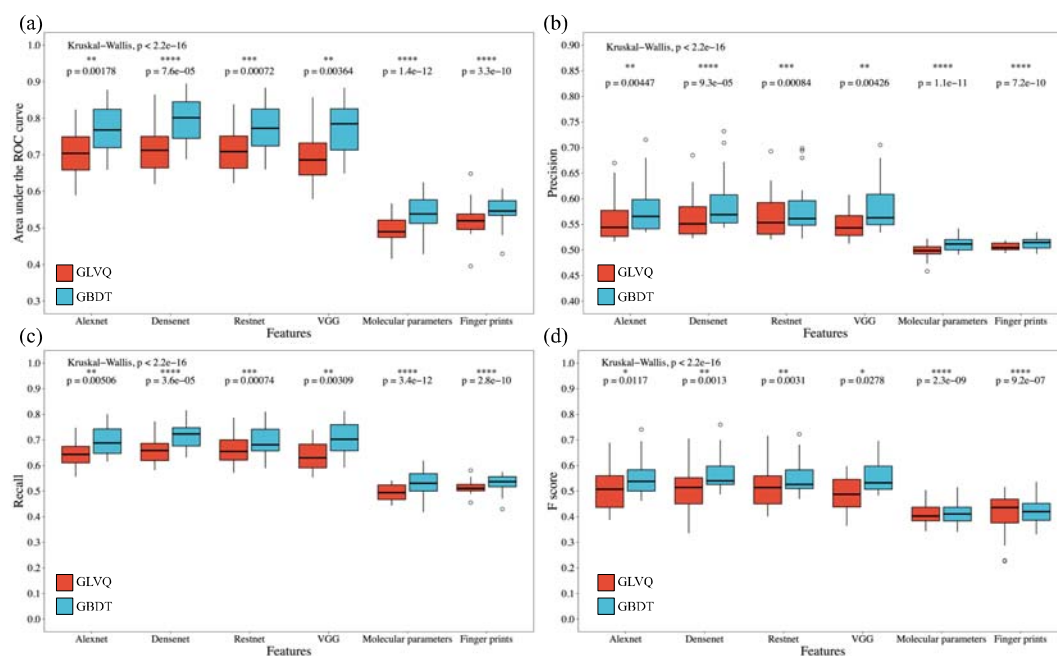


Fig. 6.

625 **Table captions**

626 Table. 1. Summary of odorant databases used in present study.

627 Table. 2. Odor cluster identification accuracy comparison of machine learning models using
628 odorant structure features.

629

630

Table 1.

Data base	# odorants	Description
Flavors and Fragrances	1026	An odor database of flavor and fragrance proposed by Sigma Aldrich.
Good Scents	3673	An odor database of flavor, fragrance, food and cosmetic industries.
OdorMapDB	321	An odor database of odorants and their olfactory bulb responses (odor maps).

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Table. 2.

Model	Features	AUC	Precision	Recall	F1 score
GLVQ	Alexnet	0.704±0.004	0.560±0.002	0.645±0.003	0.507±0.007
	Densenet	0.717±0.005	0.565±0.002	0.663±0.003	0.510±0.008
	Restnet	0.742±0.006	0.580±0.003	0.691±0.005	0.548±0.009
	VGG	0.691±0.005	0.552±0.001	0.639±0.003	0.488±0.005
	Molecular parameters	0.492±0.002	0.497±0.000	0.494±0.001	0.408±0.001
	Finger prints	0.521±0.003	0.506±0.000	0.516±0.001	0.408±0.007
GDBT	Alexnet	0.788±0.005	0.589±0.004	0.706±0.004	0.562±0.007
	Densenet	0.800±0.004	0.595±0.004	0.721±0.003	0.570±0.007
	Restnet	0.790±0.004	0.591±0.004	0.719±0.004	0.563±0.007
	VGG	0.778±0.005	0.585±0.003	0.710±0.004	0.558±0.005
	Molecular parameters	0.538±0.003	0.512±0.000	0.530±0.002	0.413±0.002
	Finger prints	0.544±0.002	0.512±0.000	0.530±0.001	0.421±0.002

634